

Skill Arena - Phase 3 Document 3

Reconciliation Engine, Reserve Calculations, Solvency Monitoring & Treasury Health Framework

Document Purpose

Define how the platform validates financial accuracy, monitors treasury health and guarantees reserve integrity.

Business Objective

Ensure every player liability is backed and every financial discrepancy is detected immediately.

Reconciliation Philosophy

The platform must continuously prove that wallet balances, treasury balances and external financial balances agree.

Core Components

Reconciliation Engine, Solvency Engine, Treasury Health Engine, Alert Engine and Audit Engine.

Reconciliation Scope

Player Wallets, Treasury Accounts, Payment Providers, Bank Accounts, Tournament Pools and Reward Pools.

Real-Time Reconciliation

Critical balances validated continuously against internal records.

Daily Reconciliation

Comprehensive reconciliation process comparing all balances and liabilities.

Reconciliation Workflow

Balance Collection → Validation → Comparison → Exception Detection → Alert Generation → Resolution Tracking.

Exception Management

Any mismatch generates an exception record requiring investigation and resolution.

Treasury Health Framework

Measures overall financial stability of the platform.

Treasury Health Metrics

Reserve Coverage, Liquidity Ratio, Liability Coverage, Withdrawal Capacity and Revenue Stability.

Solvency Monitoring

Continuously evaluates the platform's ability to meet all player obligations.

Player Liability Calculation

Total live wallet balances plus pending rewards, pending withdrawals and reserved winnings.

Reserve Coverage Calculation

Available reserves divided by total player liabilities.

Target Solvency Requirements

Treasury reserves should exceed player liabilities according to platform risk policy.

Liquidity Monitoring

Tracks immediately available funds versus short-term obligations.

Reserve Monitoring

Player Reserve, Revenue Reserve, Season Reserve, Championship Reserve, Jackpot Reserve and Emergency Reserve.

Alert Framework

Warning, Critical and Emergency severity levels.

Automated Alerts

Generated for reserve shortfalls, reconciliation failures, unusual withdrawals and treasury anomalies.

Treasury Dashboard

Displays liabilities, reserve balances, health score, solvency status and reconciliation status.

Treasury Health Score

Composite score based on liquidity, coverage, reconciliation status and operational stability.

Historical Analysis

Stores historical treasury performance and solvency trends.

Audit Integration

All reconciliation activities permanently logged and available for review.

Compliance Integration

Supports financial audits, investor reviews and regulatory examinations.

Security Requirements

Financial calculations executed exclusively on trusted server infrastructure.

Reporting Requirements

Generate daily, weekly, monthly and annual treasury reports.

Acceptance Criteria

Platform can demonstrate solvency, reserve coverage and treasury accuracy at any point in time.